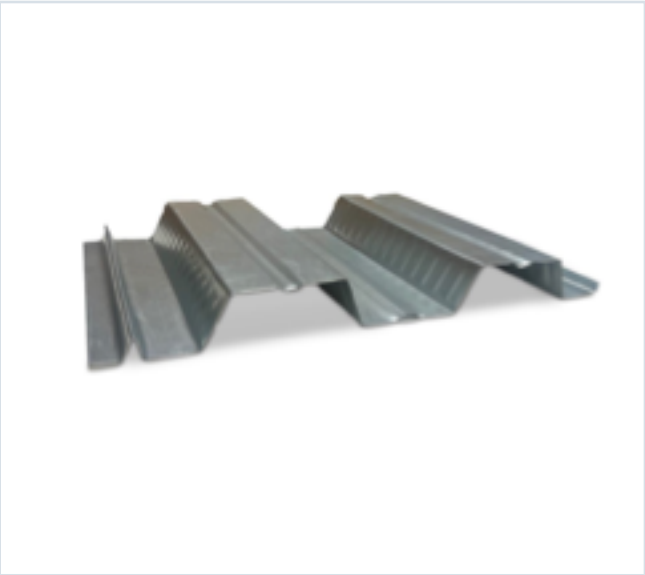


DECKING PRODUCT SUBMITTAL

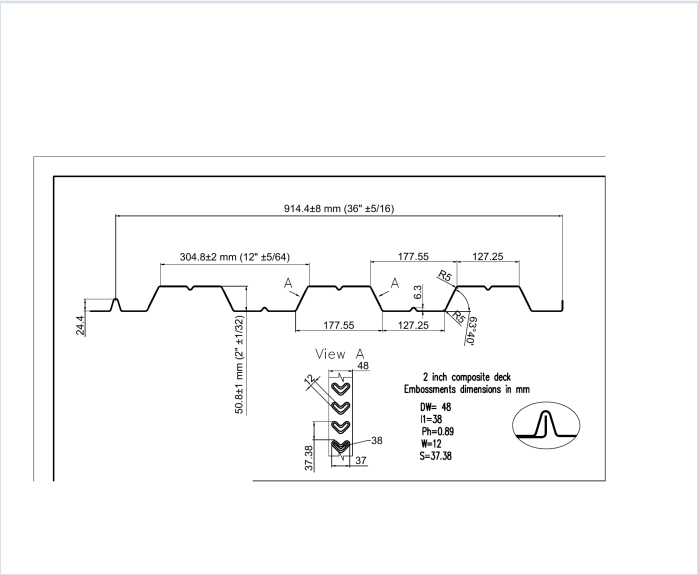
2CD-FY50-22-G40

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	22 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 22 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0295	1.6	50	0.247	0.254	7407	7595	0.311	0.290

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

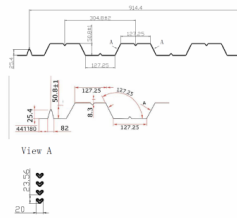
2CD-FY50-22-G40 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _s (ksi)	S _x (in ³)	S _y (in ³)	Se (in ⁴)	ASD (E _s + 16E _s)	I _x (in ⁴)	I _y (in ⁴)
22	0.0205	16	50	0.247	0.254	2427	2427	0.31	0.290
20	0.0350	19	50	0.328	0.336	3860	3860	0.393	0.373
18	0.0474	25	50	0.485	0.500	14700	14700	0.548	0.530
16	0.0598	32	50	0.646	0.664	39100	39100	0.703	0.683

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-2017, ANSI/AISC 360-2010, and ANSI/AISC 360-2005.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (V _u , kips)		Web Crippling (V _u , kips)	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	579
20	278	295	327	364	398
18	365	386	423	468	513
16	463	497	547	603	658

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-2017, ANSI/AISC 360-2010, and ANSI/AISC 360-2005.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	183	178	171	164	157	150	143	136	129	122	115
	18	273	267	258	249	240	231	222	213	204	195	186
	16	368	361	350	339	328	316	304	292	280	268	256
Double	22	131	127	121	115	109	103	97	91	85	79	73
	20	186	181	173	165	157	149	141	133	125	117	109
	18	277	271	261	251	241	231	221	211	201	191	181
	16	367	360	349	338	327	316	305	294	283	272	261
Triple	22	176	171	163	155	147	139	131	123	115	107	99
	20	243	237	228	219	210	201	192	183	174	165	156
	18	346	339	329	318	308	297	287	276	266	255	245
	16	446	438	427	416	405	394	383	372	361	350	339

Note: 1. All section properties and ASD (E_s + 16E_s) uniform loads are calculated in accordance with ANSI/AISC 360-2017, ANSI/AISC 360-2010, and ANSI/AISC 360-2005.
2. Loads shown in table are uniformly distributed superimposed loads in psi. Span lengths include center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for fixed-end moments are: Single and Two-Span: $M = \frac{wL^2}{8}$; Three-Span: $M = \frac{wL^2}{10}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	13	13	12	11	10	9	8	7	6	5	4
	18	13	13	12	11	10	9	8	7	6	5	4
	16	13	13	12	11	10	9	8	7	6	5	4
Double	22	20	19	18	17	16	15	14	13	12	11	10
	20	20	19	18	17	16	15	14	13	12	11	10
	18	20	19	18	17	16	15	14	13	12	11	10
	16	20	19	18	17	16	15	14	13	12	11	10
Triple	22	24	23	22	21	20	19	18	17	16	15	14
	20	24	23	22	21	20	19	18	17	16	15	14
	18	24	23	22	21	20	19	18	17	16	15	14
	16	24	23	22	21	20	19	18	17	16	15	14

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-22-G40 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-11"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-1"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-2"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-11"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-11"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-6"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-11"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	377	364
5	39	400	400	400	400	400	379	357
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	290	280	272	257	243	230	220	210
4.5	296	299	295	281	276	260	247	238
5	304	295	290	277	268	251	237	227
5.5	304	300	291	285	272	255	240	229
6	307	307	304	304	298	285	268	256
6.5	307	307	307	304	304	298	285	274

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	260	256	250	236	229	214	201	190
4.5	267	267	261	246	238	220	204	193
5	277	277	271	254	246	226	209	197
5.5	284	284	278	260	251	230	212	200
6	290	290	284	264	254	232	214	201
6.5	290	290	284	264	254	232	214	201